11 Lorentz Transformation of $\boldsymbol{F^{\mu\nu}}$

11.1 Deriving the Transformation of E and B

We may now derive the Lorentz transformation of **E** and **B** using $F^{\mu\nu}$. Use the standard configuration Σ, Σ'.

So:

$$F'^{\mu\nu} = \bar{\Lambda}^{\mu}{}_{\kappa}\,\bar{\Lambda}^{\nu}{}_{\lambda}\,F^{\kappa\lambda} \tag{11.1}$$

$$\Rightarrow \quad F'^{0i} = \bar{\Lambda}^{0}{}_{\kappa}\,\bar{\Lambda}^{i}{}_{\lambda}\,F^{\kappa\lambda} = \bar{\Lambda}^{0}{}_{0}\,\bar{\Lambda}^{i}{}_{j}\,F^{0j} + \bar{\Lambda}^{0}{}_{j}\,\bar{\Lambda}^{i}{}_{0}\,F^{j0} + \bar{\Lambda}^{0}{}_{j}\,\bar{\Lambda}^{i}{}_{k}\,F^{jk}. \tag{11.2}$$

11.2 Calculation of $\boldsymbol{F'^{01}}$

Consider F'^{01}:

$$\begin{aligned} F'^{01} &= \bar{\Lambda}^{0}{}_{\kappa}\,\bar{\Lambda}^{1}{}_{\lambda}\,F^{\kappa\lambda} \\ &= \bar{\Lambda}^{0}{}_{0}\,\bar{\Lambda}^{1}{}_{j}\,F^{0j} + \bar{\Lambda}^{0}{}_{j}\,\bar{\Lambda}^{1}{}_{0}\,F^{j0} + \bar{\Lambda}^{0}{}_{j}\,\bar{\Lambda}^{1}{}_{k}\,F^{jk}. \end{aligned} \tag{11.3}$$

Using the standard configuration Lorentz transformation matrix:

$$\begin{aligned} F'^{01} = {} & \cosh\phi \cdot \cosh\phi \cdot \delta^{1}{}_{j}\,F^{0j} \\ & + (-\sinh\phi)\delta^{1}{}_{j} \cdot (-\sinh\phi)\,F^{j0} \\ & + (-\sinh\phi)\delta^{1}{}_{j} \cdot \delta^{1}{}_{k} \cdot (\cosh\phi)\,F^{jk}. \end{aligned} \tag{11.4}$$

Using $F^{0j} = -F^{j0}$:

$$\begin{aligned} F'^{01} &= (\cosh^2\phi - \sinh^2\phi)F^{01} - \sinh\phi\cosh\phi\,F^{11} \\ &= 1 \times F^{01} - \cancelto{0}{F^{11}}. \end{aligned} \tag{11.5}$$

So:

$$F'^{01} = F^{01} \tag{11.6}$$

$$\Rightarrow \quad \boxed{E'_x = E_x} \tag{11.7}$$

The electric field in the direction of motion is invariant.

11.3 Calculation of $\boldsymbol{F'^{02}}$

$$\begin{aligned} F'^{02} = {} & \cosh\phi \times 1 \times F^{02} \\ & + (-\sinh\phi) \times 0 \times F^{20} \\ & + (-\sinh\phi) \times 1 \times F^{12} \end{aligned} \tag{11.8}$$

$$\Rightarrow \quad E'_y = \cosh\phi\,E_y - \sinh\phi\,B_z \tag{11.9}$$

(Using $F'^{02} = E'_y$ etc. and $F^{12} = \varepsilon^{123}B_3 = B_z$.)

11.4 Complete Field Transformations

Similarly we find:

$$E'_z = \cosh\phi\, E_z + \sinh\phi\, B_y \tag{11.10}$$

$$B'_x = B_x \tag{11.11}$$

$$B'_y = \cosh\phi\, B_y + \sinh\phi\, E_z \tag{11.12}$$

$$B'_z = \cosh\phi\, B_z - \sinh\phi\, E_y \tag{11.13}$$

We can simplify these results by dividing $\mathbf{E}$ and $\mathbf{B}$ into components parallel ($\|$) to $\mathbf{v}$, i.e. $\mathbf{E}^L$ and $\mathbf{B}^L$ (L = longitudinal), and perpendicular ($\perp$) to $\mathbf{v}$, i.e. $\mathbf{E}^T$ and $\mathbf{B}^T$ (T = transverse).

$$\boxed{\mathbf{E}^{L'} = \mathbf{E}^L} \qquad \boxed{\mathbf{B}^{L'} = \mathbf{B}^L} \tag{11.14}$$

And letting

$$\hat{\boldsymbol{\beta}} = \frac{\mathbf{v}}{\mathrm{c}} / \left|\frac{\mathbf{v}}{\mathrm{c}}\right|$$

then

$$\begin{aligned}\mathbf{E}^{T'} &= \cosh\phi\, \mathbf{E}^T + \sinh\phi\, \hat{\boldsymbol{\beta}} \times \mathbf{B}^T \\ \mathbf{B}^{T'} &= \cosh\phi\, \mathbf{B}^T - \sinh\phi\, \hat{\boldsymbol{\beta}} \times \mathbf{E}^T\ .\end{aligned}$$

11.5 Invariants of $\boldsymbol{F^{\mu\nu}}$

We found for 4-vectors that Lorentz invariants—such as $p^\mu p_\mu = -(m_0\mathrm{c})^2$—are important.

What are the invariants of $F^{\mu\nu}$?

(1) First Invariant: It is obvious that

$$F^{\mu\nu}F_{\mu\nu} = I_1 \tag{11.15}$$

is invariant.

$$\begin{aligned}F^{\mu\nu}F_{\mu\nu} &= F^{0i}F_{0i} + F^{i0}F_{i0} + F^{ij}F_{ij} \\ &= -E^2 - E^2 + \varepsilon^{ijk}\varepsilon_{ij\ell}B_kB^\ell \quad (\text{using } F^{0i} = E_i,\ F_{0i} = -E_i) \\ &= -2E^2 + 2\delta^k_\ell B_kB^\ell \\ &= -2(E^2 - B^2).\end{aligned} \tag{11.16}$$

$$\boxed{I_1 = F^{\mu\nu}F_{\mu\nu} = -2(E^2 - B^2)} \tag{11.17}$$

11.6 The 4D Levi-Civita Symbol

Given for 3d one may construct the invariants, under rotation, out of 3-vectors or tensors

$$a_i b_i \quad \text{and} \quad \varepsilon_{ijk} a_i b_j c_k \quad \text{and} \quad \varepsilon_{ijk} a_i T_{jk}$$

it is perhaps not surprising that to form the other invariant for $F^{\mu\nu}$ requires the counterpart to ε_{ijk}.

The 4D Levi-Civita symbol is defined:

$$\varepsilon_{\kappa\lambda\mu\nu} = \begin{cases} +1 & \text{if } \kappa\lambda\mu\nu \text{ is an } \textbf{even} \text{ permutation of } 0123 \\ -1 & \text{if } \kappa\lambda\mu\nu \text{ is an } \textbf{odd} \text{ permutation of } 0123 \\ 0 & \text{otherwise} \end{cases} \tag{11.18}$$

Notes: **(i)** Raising indices:

$$\varepsilon^{\kappa\lambda\mu\nu} = \eta^{\kappa\alpha}\eta^{\lambda\beta}\eta^{\mu\gamma}\eta^{\nu\delta}\varepsilon_{\alpha\beta\gamma\delta}. \tag{11.19}$$

Pick $\kappa\lambda\mu\nu = 0123$. Then:

$$\begin{aligned} \varepsilon^{0123} &= \eta^{0\alpha}\eta^{1\beta}\eta^{2\gamma}\eta^{3\delta}\varepsilon_{\alpha\beta\gamma\delta} \\ &= -\delta^\alpha_0\delta^\beta_1\delta^\gamma_2\delta^\delta_3\varepsilon_{\alpha\beta\gamma\delta} \\ &= -\varepsilon_{0123} \end{aligned} \tag{11.20}$$

and the $-$ sign is valid for **any choice** of $\kappa\lambda\mu\nu$.

Consequence: $\varepsilon_{0ijk} = \varepsilon_{ijk}$, but $\varepsilon^{0ijk} = -\varepsilon_{ijk}$.

(ii) $\varepsilon_{\kappa\lambda\mu\nu}$ is a **pseudotensor**, using logic as in problem (2.10) for ε_{ijk}:

$$\varepsilon_{\kappa\lambda\mu\nu} = |\Lambda|\, \Lambda_\kappa{}^\alpha \Lambda_\lambda{}^\beta \Lambda_\mu{}^\gamma \Lambda_\nu{}^\delta \varepsilon_{\alpha\beta\gamma\delta}. \tag{11.21}$$

(iii) Note: **cyclic permutations are odd in 4D**:

$$0123 \to 1023 \to 1203 \to 1230 \tag{11.22}$$

requires 3 adjacent transpositions.

$$\Rightarrow \quad \varepsilon_{1230} = -\varepsilon_{0123}. \tag{11.23}$$

(iv) Contraction identity (proven from the counterpart to the identity in problem (2.13)):

$$\varepsilon^{\alpha\beta\gamma\delta}\varepsilon_{\mu\nu\gamma\delta} = 2(\delta^\alpha_\mu\delta^\beta_\nu - \delta^\alpha_\nu\delta^\beta_\mu)\ . \tag{11.24}$$

Identities with more indices contracted are:

$$\begin{aligned} \varepsilon^{\alpha\beta\gamma\delta}\varepsilon_{\mu\beta\gamma\delta} &= 2(\delta^\alpha_\mu \underbrace{\delta^\beta_\beta}_{=4} - \delta^\alpha_\beta\delta^\beta_\mu) \\ &= 2(4\delta^\alpha_\mu - \delta^\alpha_\mu) \\ &= 6\,\delta^\alpha_\mu \\ \Rightarrow \quad \varepsilon^{\alpha\beta\gamma\delta}\varepsilon_{\alpha\beta\gamma\delta} &= 24\ . \quad \text{(All indices contracted)} \end{aligned}$$

11.7 The Dual Tensor ${}^*\boldsymbol{F^{\mu\nu}}$

By analogy with 3d Cartesian tensors, we may define 4d, relativistic, duals using the 4d Levi-Civita symbol. This will result in a neat expression for two of Maxwell's equations in covariant form.

We define the **dual** of $F^{\mu\nu}$:

$$\begin{aligned} {}^*F^{\mu\nu} &= \frac{1}{2}\varepsilon^{\mu\nu\kappa\lambda}F_{\kappa\lambda} \\ &= \frac{1}{2}\varepsilon^{\mu\nu\kappa\lambda}(\partial_\kappa A_\lambda - \partial_\lambda A_\kappa) \\ &= \varepsilon^{\mu\nu\kappa\lambda}\partial_\kappa A_\lambda . \end{aligned} \tag{11.25}$$

Note: ${}^*F^{\mu\nu}$ is a rank-2 **pseudotensor**, due to ε.

Components of ${}^*\boldsymbol{F^{\mu\nu}}$
Again we divide the consideration into time-space components and purely spatial components, anticipating a division between **E** and **B**.

$$\begin{aligned} {}^*F^{0i} &= \varepsilon^{0ijk}\partial_j A_k \quad \text{(first index = 0 so rest are spatial)} \\ &= -\varepsilon^{ijk}\partial_j A_k \\ &= -B^i \ . \\ \text{and} \quad {}^*F^{ij} &= \varepsilon^{ij0k}\partial_0 A_k + \varepsilon^{ijk0}\partial_k A_0 \\ &= \varepsilon^{0ijk}\partial_0 A_k - \varepsilon^{0ijk}\partial_k A_0 \quad \text{(2 AT's, 3 AT's)} \\ &= \varepsilon^{0ijk}\partial_0 A_k + \varepsilon^{0ijk}\partial_k A^0 \quad (A_0 = -A^0 = -\varphi) \\ &= -\varepsilon^{ijk}\left(\frac{1}{\mathrm{c}}\frac{\partial \mathbf{A}}{\partial t} + \partial_k\varphi\right)_k \quad (\text{As } A^0 = \varphi,\ \varepsilon^{0ijk} = -\varepsilon^{ijk}) \\ {}^*F^{ij} &= +\varepsilon^{ijk}E_k \ . \end{aligned}$$

So $+E_k$ is the dual of ${}^*F^{ij}$:

$$E^i = +\frac{1}{2}\varepsilon^{ijk}\,{}^*F_{jk}. \tag{11.26}$$

$$\Rightarrow \quad \boxed{{}^*F^{\mu\nu} = \begin{pmatrix} 0 & -B_x & -B_y & -B_z \\ +B_x & 0 & +E_z & -E_y \\ +B_y & -E_z & 0 & +E_x \\ +B_z & +E_y & -E_x & 0 \end{pmatrix}} \tag{11.27}$$

So $\mathbf{E} \to -\mathbf{B}$, $\mathbf{B} \to +\mathbf{E}$: is termed the "**duality transformation**" (see problems (3.9) and (3.10) for exploration – including the possibility of magnetic monopoles).

11.8 Second Invariant

(2) We may define a now-obvious second invariant using the dual tensor:

$$\begin{aligned} I_2 &= {}^*F^{\mu\nu}F_{\mu\nu} \\ &= {}^*F^{0i}F_{0i} + {}^*F^{i0}F_{i0} + {}^*F^{ij}F_{ij} + \cancelto{0}{{}^*F^{00}F_{00}} \\ &= +\mathbf{B}\cdot\mathbf{E} + (+\mathbf{B})\cdot(+\mathbf{E}) + (+\varepsilon^{ijk}E_k)\varepsilon_{ij\ell}B^\ell \\ &= +2\mathbf{B}\cdot\mathbf{E} + 2\delta^k_\ell B^\ell E_k \\ &= +4\mathbf{B}\cdot\mathbf{E}\ . \end{aligned} \tag{11.28}$$

For an exploration of other potential invariants, see problem (3.4).

$$\boxed{I_2 = {}^*F^{\mu\nu}F_{\mu\nu} = 4\mathbf{B}\cdot\mathbf{E}} \tag{11.29}$$

11.9 Consequences

11.9.1 Can we choose a frame to eliminate one of E or B?

Suppose $\mathbf{B} = 0$ in some frame. Then:

(i) $E^2 - B^2 =$ invariant $\Rightarrow E > B$ in all frames.

(ii) $\mathbf{E}\cdot\mathbf{B} = 0$ in all frames (as invariant).

So when $|\mathbf{B}| \neq 0$, then $\mathbf{E} \perp \mathbf{B}$.

Thus if $\mathbf{E} \not\perp \mathbf{B}$, then we **cannot eliminate either** by a Lorentz transformation.

11.9.2 Special Case: $|\mathbf{E}| = |\mathbf{B}|$ and $\mathbf{E} \perp \mathbf{B}$

Note special case:
If in some frame $|\mathbf{E}| = |\mathbf{B}|$ and $\mathbf{E}\cdot\mathbf{B} = 0$ (so $I_1 = I_2 = 0$), then this is true in **all** frames.

Examples: **(i) Static fields** (with $E_0 = B_0$):

$$\mathbf{E} = E_0\hat{\mathbf{x}}, \qquad \mathbf{B} = B_0\hat{\mathbf{y}}. \tag{11.30}$$

(ii) Plane wave, linearly polarised:

$$\mathbf{E} = E_0\hat{\mathbf{x}}\cos(kz - \omega t) \tag{11.31}$$

$$\mathbf{B} = B_0\hat{\mathbf{y}}\cos(kz - \omega t) \tag{11.32}$$

with $E_0 = B_0$. I.e. plane waves are transverse in **all** frames.

Problems (3.5-7) explore the use of invariants in a variety of contexts.